

Ashish Yadav

Backend Engineer · Distributed Systems & AI Infrastructure

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SUMMARY

Backend engineer specializing in serverless architecture and AI-integrated systems - with hands-on experience across distributed systems, multi-tenant SaaS, and cloud infrastructure. Ships end-to-end across FinTech, AI, and SaaS domains, and actively contributes to open source.

EXPERIENCE

ASOasis Tech

Aug. 2025 – Present

Software Developer Intern

Remote

- Designed and deployed a production-grade serverless backend (Go + AWS Lambda) serving **100K+ downloads** with real-time AI chat at **sub-200ms latency** - integrating OpenAI GPT-4.1, a Gorse recommendation engine (Redis + MySQL), and a sigmoid-based match probability formula across 5 weighted engagement signals
- Architected a fault-tolerant distributed content automation system with a 3-stage pipeline (Source → Generator → Destination), poll-based scheduler via EventBridge + DynamoDB conditional locking (10-min TTL), and 2–8 parallel workers orchestrating multimodal AI output across tenant-configurable frequencies
- Delivered a multi-tenant SaaS workflow platform with full lifecycle orchestration (create → validate → activate → run), concurrent execution engine, passwordless auth (OTP + magic link + Google OAuth), and custom CloudWatch observability - production-ready from day one
- Developed a real-time product analytics backend (AWS Lambda + BigQuery) surfacing DAU, funnel conversion rates, and behavioral event-coincidence metrics; introduced a 5-min TTL cache slashing BigQuery call volume by **~95%** with injection-safe parameterized SQL across 90-day query windows

Freelance Software Developer

Nov. 2024 – Jul. 2025

Client-Based Projects

Remote

- Shipped Swipe It - an AI-powered recruitment platform using vector embeddings to elevate candidate-role match precision by **80%** over conventional keyword search; deployed end-to-end to production for a recruitment startup
- Launched an ERP backend with an admin-controlled Formula Engine achieving **100% stock reconciliation accuracy**; automated order-to-production workflows via intelligent email parsing, reducing operational overhead by **90%**
- Owned complete product lifecycles across **3 domains** - FinTech, manufacturing, and recruitment - driving requirements gathering, system design, and cloud deployment independently for each client engagement

PROJECTS

Mini-Dropbox

Nov. 2024

Go · REST API · HTTP/2 · sync.Pool · Concurrency · Distributed Systems

[GitHub](#)

- Forged a distributed file storage system in Go with master-node coordination and two storage nodes, backed by content-addressed storage via MD5 hashing and a RESTful HTTP API for file registration, deduplication, and cross-node retrieval
- Engineered sync.Pool buffer pooling, RWMutex-protected ops, and an in-memory existence cache - achieving **4x concurrent throughput** (500→2,000 req/s), **43% memory reduction** (150MB→85MB), and **55% faster retrieval** (45ms→20ms)

ASHOS - AI-Powered CLI Personal OS

Sept. 2024

Go · OpenAI · RAG · SQLite · Vector Search · Concurrency

[GitHub](#)

- Constructed a production-grade, event-driven CLI OS in Go with Composition Root DI and interface-driven storage - enabling full infrastructure swap in 1 line; enforced sync.RWMutex safety across 6+ concurrent modules
- Crafted a full RAG engine from scratch: OpenAI embeddings → hybrid vector search (SQLite-vec + custom Cosine Similarity) → GPT response spanning 4+ data sources; exposed 4 AI productivity commands backed by an async ingestion loop

EDUCATION

K.R. Mangalam University

2023 – 2027

B.Tech in Computer Science and Engineering

- Relevant Coursework:** Data Structures, Algorithms, Object-Oriented Design, Database Management Systems, Operating Systems, Computer Networks, Complexity Analysis

SKILLS

Languages: Go (net/http, Gin, GORM), TypeScript, JavaScript (ES6+), Python

Backend & Infrastructure: Node.js, REST APIs, AWS (Lambda, S3, DynamoDB, EventBridge, SQS, CloudWatch), Docker, Firebase, Git

Frontend & Mobile: React.js, Next.js, Flutter

Databases & AI: PostgreSQL, MongoDB, Redis, BigQuery, SQLite, OpenAI API, Vector Databases (pgvector, SQLite-vec)